Shubham Bhardwaj

PhD Student in Theoretical Computer Science

School of Technology and Computer Science
Tata Institute of Fundamental Research, Mumbai
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RESEARCH INTERESTS

Complexity Theory, Randomised Algorithms, Machine Learning, AI Safety.

EDUCATION

PhD in Computer Science 2024 – present
School of Technology and Computer Science, Tata Institute of Fundamental Research, Mumbai
Advisor: Ramprasad Saptharishi

M.Math. 2021 – 2023
Indian Statistical Institute, Bangalore

B.Math. (Hons.) 2018 – 2021
Indian Statistical Institute, Bangalore

RESEARCH EXPERIENCE

Research Assistant 2023 – 2024
Worked with Sandeep Sen on designing instance-optimal algorithms.

Project in Quantitative Finance Jan – May 2023
Least Squares Monte Carlo Method for Option Pricing
With Rituparna Sen, Indian Statistical Institute, Bangalore

PAPERS

PREPRINT · 2026

Hitting Sets for Polynomials with Small Partial Derivative Spaces
with Ramprasad Saptharishi
arXiv:2609.34436 ECCC TR26-217

FSTTCS 2026

Unconditional Separation Between LL and CLL
To appear in the 46th IARCS Annual Conference on Foundations of Software Technology and Theoretical Computer Science (FSTTCS 2026).
arXiv:2610.15554 ECCC TR26-252